

Probability & Random Processes EE2011

Department of Electrical Engineering

**Gamma, Exponential, Chi Squared, Beta,
lognormal, Weibull Distributions**

NATIONAL UNIVERSITY OF COMPUTER & EMERGING
SCIENCES, LAHORE CAMPUS



6.6 Gamma and Exponential Distributions

Although the normal distribution can be used to solve many problems in engineering and science, there are still numerous situations that require different types of density functions.

Two such density functions, the **gamma** and **exponential distributions**, are discussed in this section.

It turns out that the exponential distribution is a special case of the gamma distribution.

Both find a large number of applications.

The exponential and gamma distributions play an important role in both queuing theory and reliability problems.

Time between arrivals at service facilities and time to failure of component parts and electrical systems often are nicely modeled by the exponential distribution.

The relationship between the gamma and the exponential allows the gamma to be used in similar types of problems.

More details and illustrations will be supplied later in the section.



The gamma distribution derives its name from the well-known **gamma function**, studied in many areas of mathematics.

Before we proceed to the gamma distribution, let us review this function and some of its important properties.

Definition 6.2:

The **gamma function** is defined by

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx, \quad \text{for } \alpha > 0.$$

The following are a few simple properties of the gamma function.

(a) $\Gamma(n) = (n-1)(n-2) \cdots (1)\Gamma(1)$, for a positive integer n .

To see the proof, integrating by parts with $u = x^{\alpha-1}$ and $dv = e^{-x} dx$, we obtain

$$\Gamma(\alpha) = -e^{-x} x^{\alpha-1} \Big|_0^{\infty} + \int_0^{\infty} e^{-x} (\alpha-1) x^{\alpha-2} dx = (\alpha-1) \int_0^{\infty} x^{\alpha-2} e^{-x} dx,$$

for $\alpha > 1$, which yields the recursion formula

$$\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1).$$

The result follows after repeated application of the recursion formula.

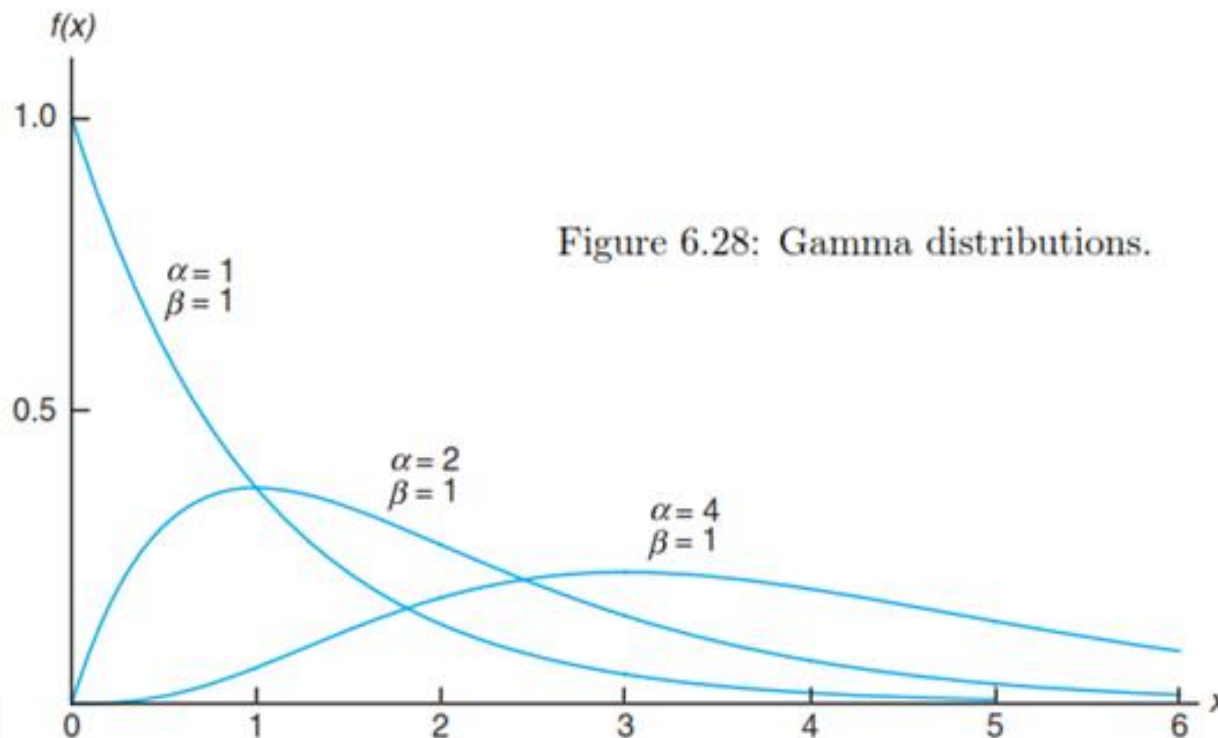


Gamma Distribution

The continuous random variable X has a **gamma distribution**, with parameters α and β , if its density function is given by

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, & x > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

where $\alpha > 0$ and $\beta > 0$.



Exponential Distribution

The continuous random variable X has an **exponential distribution**, with parameter β , if its density function is given by

$$f(x; \beta) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & x > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

where $\beta > 0$.

The following theorem and corollary give the mean and variance of the gamma and exponential distributions.

Theorem 6.4: The mean and variance of the gamma distribution are

$$\mu = \alpha\beta \text{ and } \sigma^2 = \alpha\beta^2.$$

The proof of this theorem is found in Appendix A.26.

Corollary 6.1: The mean and variance of the exponential distribution are

$$\mu = \beta \text{ and } \sigma^2 = \beta^2.$$



Relationship to the Poisson Process

We shall pursue applications of the exponential distribution and then return to the gamma distribution.

The most important applications of the exponential distribution are situations where the Poisson process applies (see Section 5.5).

The reader should recall that the Poisson process allows for the use of the discrete distribution called the Poisson distribution.

Recall that the Poisson distribution is used to compute the probability of specific numbers of “events” during a particular *period of time or span of space*.

In many applications, the time period or span of space is the random variable.

For example, an industrial engineer may be interested in modeling the time T between arrivals at a congested intersection during rush hour in a large city.

An arrival represents the Poisson event.



The relationship between the exponential distribution (often called the negative exponential) and the Poisson process is quite simple.

In Chapter 5, the Poisson distribution was developed as a single-parameter distribution with parameter λ , where λ may be interpreted as the mean number of events *per unit “time.”*

Consider now the *random variable* described by the time required for the first event to occur.

Using the Poisson distribution, we find that the probability of no events occurring in the span up to time t is given by

$$p(0; \lambda t) = \frac{e^{-\lambda t} (\lambda t)^0}{0!} = e^{-\lambda t}.$$

We can now make use of the above and let X be the time to the first Poisson event.

The probability that the length of time until the first event will exceed x is the same as the probability that no Poisson events will occur in x .

The latter, of course, is given by $e^{-\lambda x}$.



As a result, $P(X > x) = e^{-\lambda x}$.

Thus, the cumulative distribution function for X is given by

$$P(0 \leq X \leq x) = 1 - e^{-\lambda x}.$$

Now, in order that we may recognize the presence of the exponential distribution, we differentiate the cumulative distribution function above to obtain the density function

$$f(x) = \lambda e^{-\lambda x},$$

which is the density function of the exponential distribution with $\lambda = 1/\beta$.



Applications of the Exponential and Gamma Distributions

In the foregoing, we provided the foundation for the application of the exponential distribution in “time to arrival” or time to Poisson event problems.

We will illustrate some applications here and then proceed to discuss the role of the gamma distribution in these modeling applications.

Notice that the mean of the exponential distribution is the parameter β , the reciprocal of the parameter in the Poisson distribution.

The reader should recall that it is often said that the Poisson distribution has no memory, implying that occurrences in successive time periods are independent.

The important parameter β is the mean time between events.

In reliability theory, where equipment failure often conforms to this Poisson process, β is called **mean time between failures**.

Many equipment breakdowns do follow the Poisson process, and thus the exponential distribution does apply.

Other applications include survival times in biomedical experiments and computer response time.



Example 6.17: Suppose that a system contains a certain type of component whose time, in years, to failure is given by T . The random variable T is modeled nicely by the exponential distribution with mean time to failure $\beta = 5$. If 5 of these components are installed in different systems, what is the probability that at least 2 are still functioning at the end of 8 years?

Solution: The probability that a given component is still functioning after 8 years is given by

$$P(T > 8) = \frac{1}{5} \int_8^{\infty} e^{-t/5} dt = e^{-8/5} \approx 0.2.$$

Let X represent the number of components functioning after 8 years. Then using the binomial distribution, we have

$$P(X \geq 2) = \sum_{x=2}^5 b(x; 5, 0.2) = 1 - \sum_{x=0}^1 b(x; 5, 0.2) = 1 - 0.7373 = 0.2627.$$



The Memoryless Property and Its Effect on the Exponential Distribution

The types of applications of the exponential distribution in reliability and component or machine lifetime problems are influenced by the **memoryless** (or lack-of-memory) **property** of the exponential distribution.

For example, in the case of, say, an electronic component where lifetime has an exponential distribution, the probability that the component lasts, say, t hours, that is, $P(X \geq t)$, is the same as the conditional probability

$$P(X \geq t_0 + t \mid X \geq t_0).$$

So if the component “makes it” to t_0 hours, the probability of lasting an additional t hours is the same as the probability of lasting t hours.

There is no “punishment” through wear that may have ensued for lasting the first t_0 hours.

Thus, the exponential distribution is more appropriate when the memoryless property is justified.

But if the failure of the component is a result of gradual or slow wear (as in mechanical wear), then the exponential does not apply and either the gamma or the Weibull distribution (Section 6.10) may be more appropriate.



The importance of the gamma distribution lies in the fact that it defines a family of which other distributions are special cases. But the gamma itself has important applications in waiting time and reliability theory.

Whereas the exponential distribution describes the time until the occurrence of a Poisson event (or the time between Poisson events), the time (or space) occurring until a *specified number of Poisson events occur* is a random variable whose density function is described by the gamma distribution.

This specific number of events is the parameter α in the gamma density function.

Thus, it becomes easy to understand that when $\alpha = 1$, the special case of the exponential distribution occurs.

The gamma density can be developed from its relationship to the Poisson process in much the same manner as we developed the exponential density.

The details are left to the reader.

The following is a numerical example of the use of the gamma distribution in a waiting-time application.



Example 6.18: Suppose that telephone calls arriving at a particular switchboard follow a Poisson process with an average of 5 calls coming per minute. What is the probability that up to a minute will elapse by the time 2 calls have come in to the switchboard?

Solution: The Poisson process applies, with time until 2 Poisson events following a gamma distribution with $\beta = 1/5$ and $\alpha = 2$. Denote by X the time in minutes that transpires before 2 calls come. The required probability is given by

$$P(X \leq 1) = \int_0^1 \frac{1}{\beta^2} x e^{-x/\beta} dx = 25 \int_0^1 x e^{-5x} dx = 1 - e^{-5}(1 + 5) = 0.96. \quad \text{J}$$



While the origin of the gamma distribution deals in time (or space) until the occurrence of α Poisson events, there are many instances where a gamma distribution works very well even though there is no clear Poisson structure.

This is particularly true for **survival time** problems in both engineering and biomedical applications.



Example 6.19: In a biomedical study with rats, a dose-response investigation is used to determine the effect of the dose of a toxicant on their survival time. The toxicant is one that is frequently discharged into the atmosphere from jet fuel. For a certain dose of the toxicant, the study determines that the survival time, in weeks, has a gamma distribution with $\alpha = 5$ and $\beta = 10$. What is the probability that a rat survives no longer than 60 weeks?

Solution: Let the random variable X be the survival time (time to death). The required probability is

$$P(X \leq 60) = \frac{1}{\beta^5} \int_0^{60} \frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(5)} dx.$$

The integral above can be solved through the use of the **incomplete gamma function**, which becomes the cumulative distribution function for the gamma distribution. This function is written as

$$F(x; \alpha) = \int_0^x \frac{y^{\alpha-1} e^{-y}}{\Gamma(\alpha)} dy.$$

If we let $y = x/\beta$, so $x = \beta y$, we have

$$P(X \leq 60) = \int_0^6 \frac{y^4 e^{-y}}{\Gamma(5)} dy,$$

which is denoted as $F(6; 5)$ in the table of the incomplete gamma function in Appendix A.23. Note that this allows a quick computation of probabilities for the gamma distribution. Indeed, for this problem, the probability that the rat survives no longer than 60 days is given by

$$P(X \leq 60) = F(6; 5) = 0.715.$$



Example 6.20: It is known, from previous data, that the length of time in months between customer complaints about a certain product is a gamma distribution with $\alpha = 2$ and $\beta = 4$. Changes were made to tighten quality control requirements. Following these changes, 20 months passed before the first complaint. Does it appear as if the quality control tightening was effective?

Solution: Let X be the time to the first complaint, which, under conditions prior to the changes, followed a gamma distribution with $\alpha = 2$ and $\beta = 4$. The question centers around how rare $X \geq 20$ is, given that α and β remain at values 2 and 4, respectively. In other words, under the prior conditions is a “time to complaint” as large as 20 months reasonable? Thus, following the solution to Example 6.19,

$$P(X \geq 20) = 1 - \frac{1}{\beta^\alpha} \int_0^{20} \frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha)} dx.$$

Again, using $y = x/\beta$, we have

$$P(X \geq 20) = 1 - \int_0^5 \frac{ye^{-y}}{\Gamma(2)} dy = 1 - F(5; 2) = 1 - 0.96 = 0.04,$$

where $F(5; 2) = 0.96$ is found from Table A.23.

As a result, we could conclude that the conditions of the gamma distribution with $\alpha = 2$ and $\beta = 4$ are not supported by the data that an observed time to complaint is as large as 20 months. Thus, it is reasonable to conclude that the quality control work was effective.



Example 6.21: Consider Exercise 3.31 on page 94. Based on extensive testing, it is determined that the time Y in years before a major repair is required for a certain washing machine is characterized by the density function

$$f(y) = \begin{cases} \frac{1}{4}e^{-y/4}, & y \geq 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Note that Y is an exponential random variable with $\mu = 4$ years. The machine is considered a bargain if it is unlikely to require a major repair before the sixth year. What is the probability $P(Y > 6)$? What is the probability that a major repair is required in the first year?

Solution: Consider the cumulative distribution function $F(y)$ for the exponential distribution,

$$F(y) = \frac{1}{\beta} \int_0^y e^{-t/\beta} dt = 1 - e^{-y/\beta}.$$

Then

$$P(Y > 6) = 1 - F(6) = e^{-3/2} = 0.2231.$$

Thus, the probability that the washing machine will require major repair after year six is 0.223. Of course, it will require repair before year six with probability 0.777. Thus, one might conclude the machine is not really a bargain. The probability that a major repair is necessary in the first year is

$$P(Y < 1) = 1 - e^{-1/4} = 1 - 0.779 = 0.221.$$



6.7 Chi-Squared Distribution

Another very important special case of the gamma distribution is obtained by letting $\alpha = v/2$ and $\beta = 2$, where v is a positive integer.

The result is called **the chi-squared distribution**.

The distribution has a single parameter, v , called the **degrees of freedom**.

Chi-Squared Distribution

The continuous random variable X has a **chi-squared distribution**, with v **degrees of freedom**, if its density function is given by

$$f(x; v) = \begin{cases} \frac{1}{2^{v/2}\Gamma(v/2)} x^{v/2-1} e^{-x/2}, & x > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

where v is a positive integer.



The chi-squared distribution plays a vital role in statistical inference.

It has considerable applications in both methodology and theory.

While we do not discuss applications in detail in this chapter, it is important to understand that Chapters 8, 9, and 16 contain important applications.

The chi-squared distribution is an important component of statistical hypothesis testing and estimation.

Topics dealing with sampling distributions, analysis of variance, and nonparametric statistics involve extensive use of the chi-squared distribution.

Theorem 6.5:

The mean and variance of the chi-squared distribution are

$$\mu = v \text{ and } \sigma^2 = 2v.$$



6.8 Beta Distribution

An extension to the uniform distribution is a beta distribution.

Let us start by defining a **beta function**.

Definition 6.3:

A **beta function** is defined by

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}, \text{ for } \alpha, \beta > 0,$$

where $\Gamma(\alpha)$ is the gamma function.

Beta Distribution

The continuous random variable X has a **beta distribution** with parameters $\alpha > 0$ and $\beta > 0$ if its density function is given by

$$f(x) = \begin{cases} \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Note that the uniform distribution on $(0, 1)$ is a beta distribution with parameters $\alpha = 1$ and $\beta = 1$.



Theorem 6.6:

The mean and variance of a beta distribution with parameters α and β are

$$\mu = \frac{\alpha}{\alpha + \beta} \text{ and } \sigma^2 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)},$$

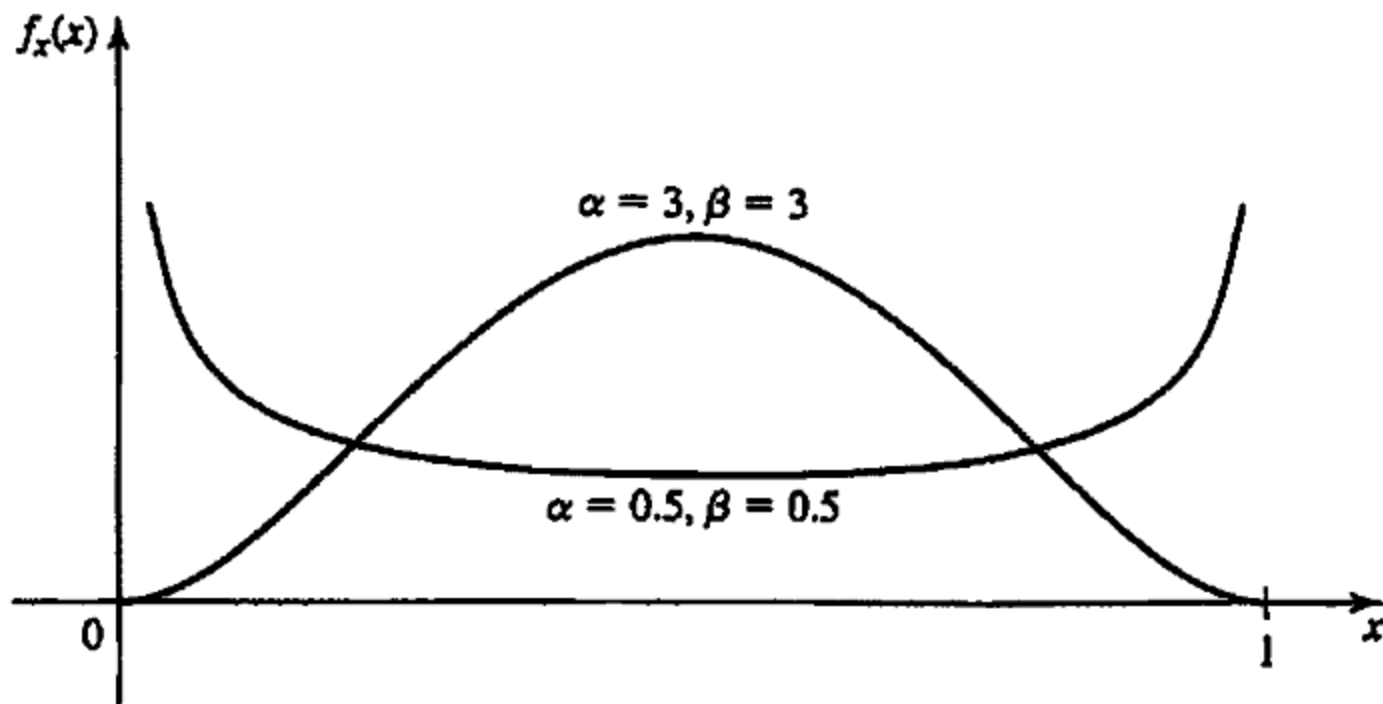
respectively.

For the uniform distribution on $(0, 1)$, the mean and variance are

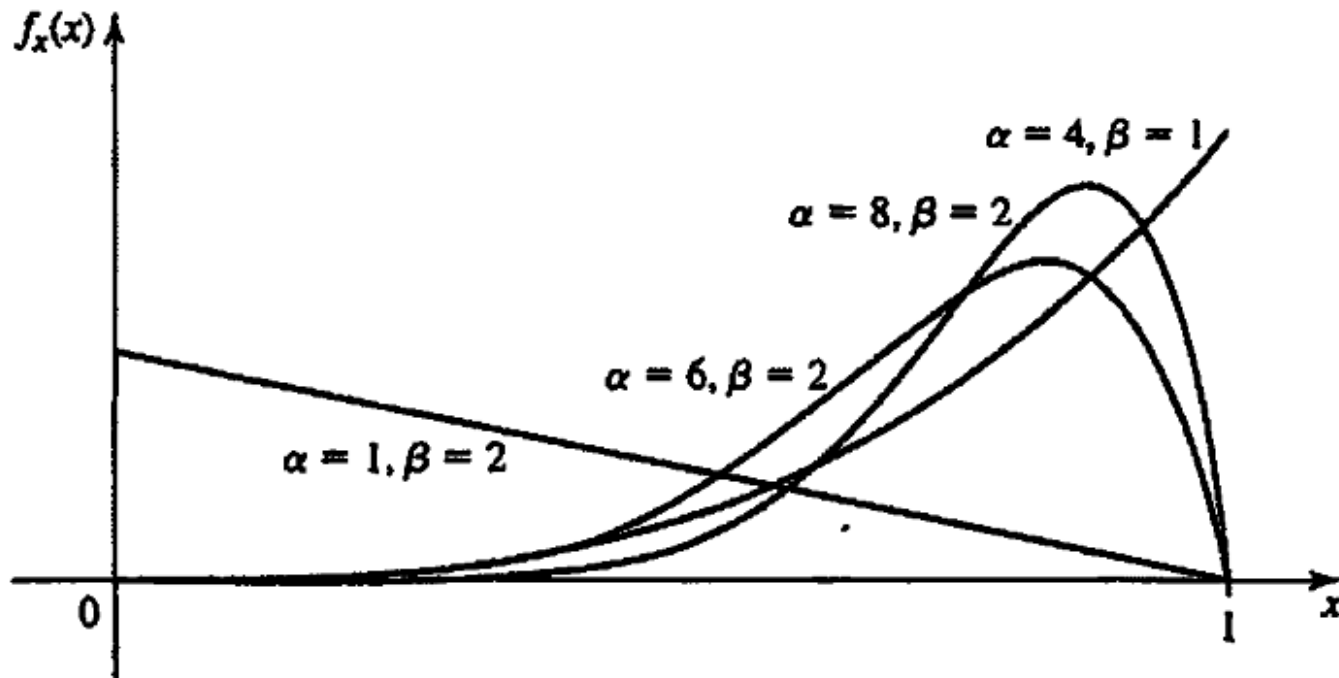
$$\mu = \frac{1}{1 + 1} = \frac{1}{2} \text{ and } \sigma^2 = \frac{(1)(1)}{(1 + 1)^2(1 + 1 + 1)} = \frac{1}{12},$$

respectively.





(a) $(\alpha, \beta) = (0.5, 0.5), (3, 3)$



(b) $(\alpha, \beta) = (1, 2), (4, 1), (6, 2), (8, 2)$

6.9 Lognormal Distribution

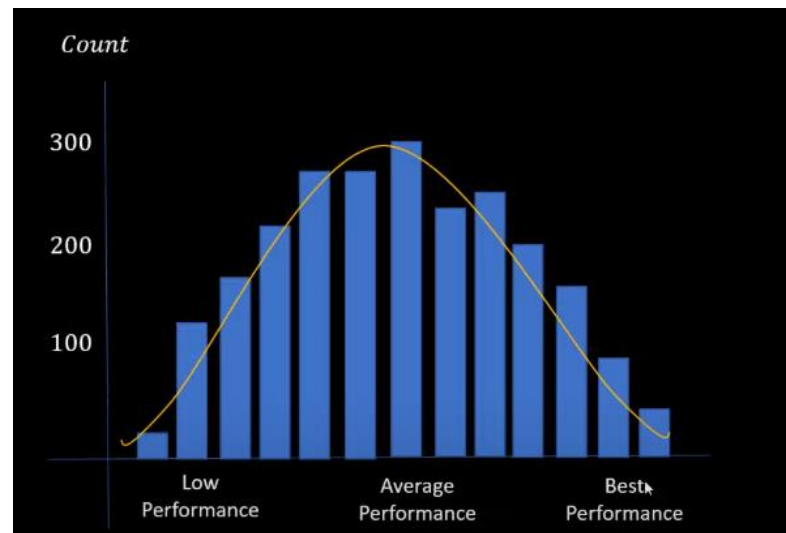
The lognormal distribution is used for a wide variety of applications. The distribution applies in cases where a natural log transformation results in a normal distribution.

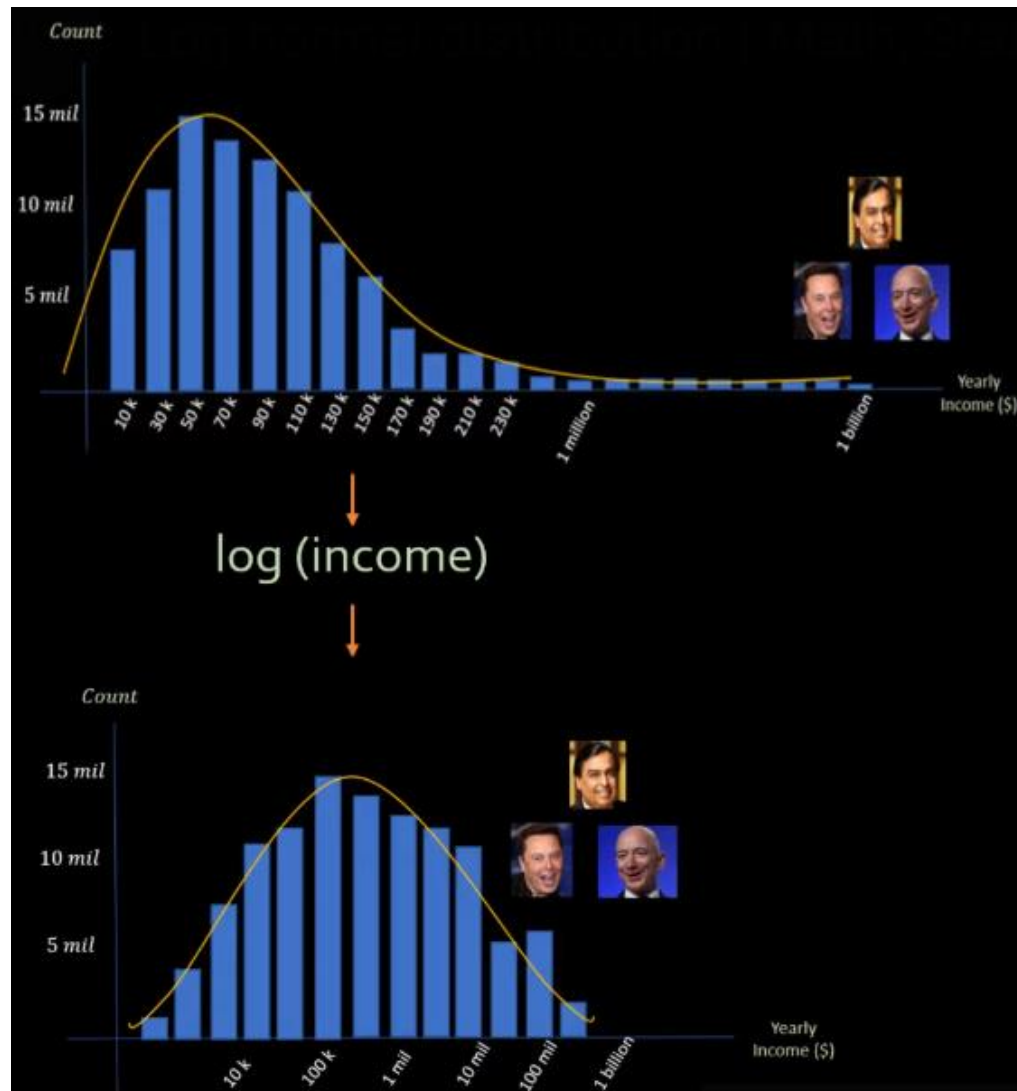
Lognormal Distribution

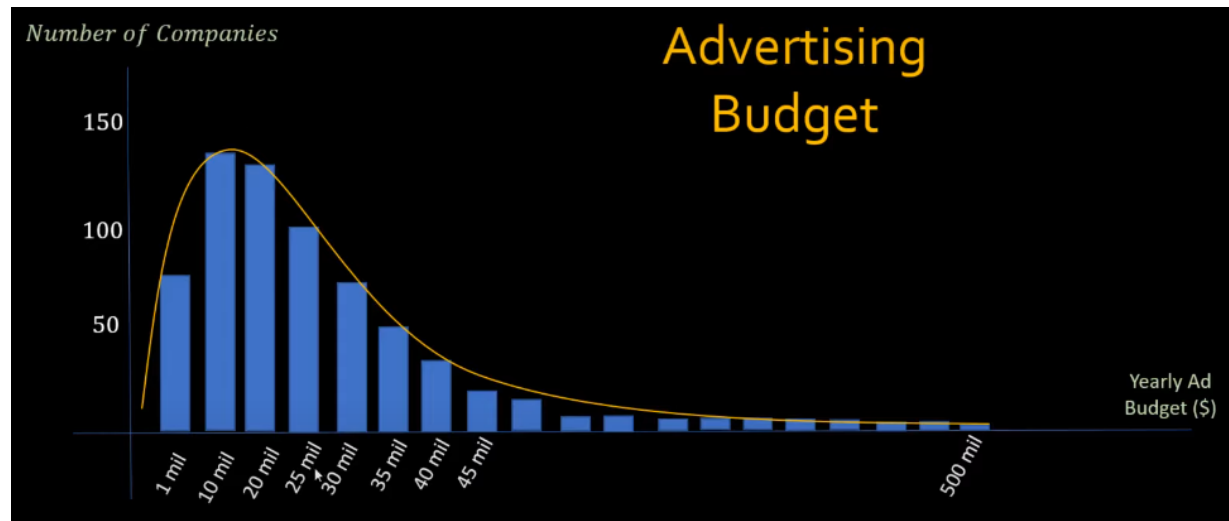
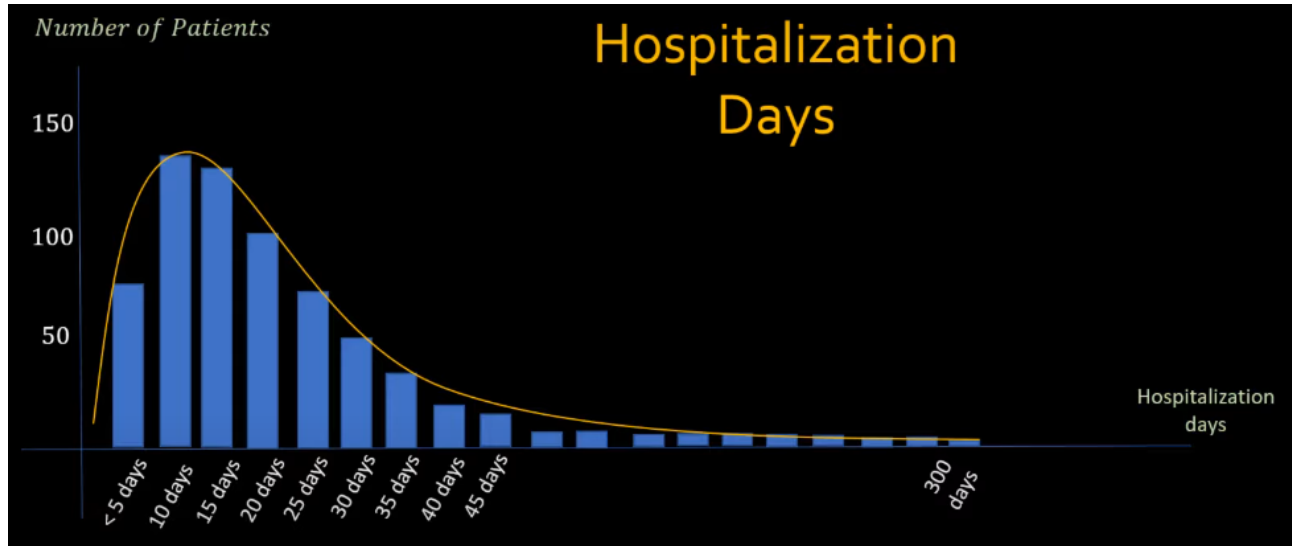
The continuous random variable X has a **lognormal distribution** if the random variable $Y = \ln(X)$ has a normal distribution with mean μ and standard deviation σ . The resulting density function of X is

$$f(x; \mu, \sigma) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma x} e^{-\frac{1}{2\sigma^2} [\ln(x) - \mu]^2}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$









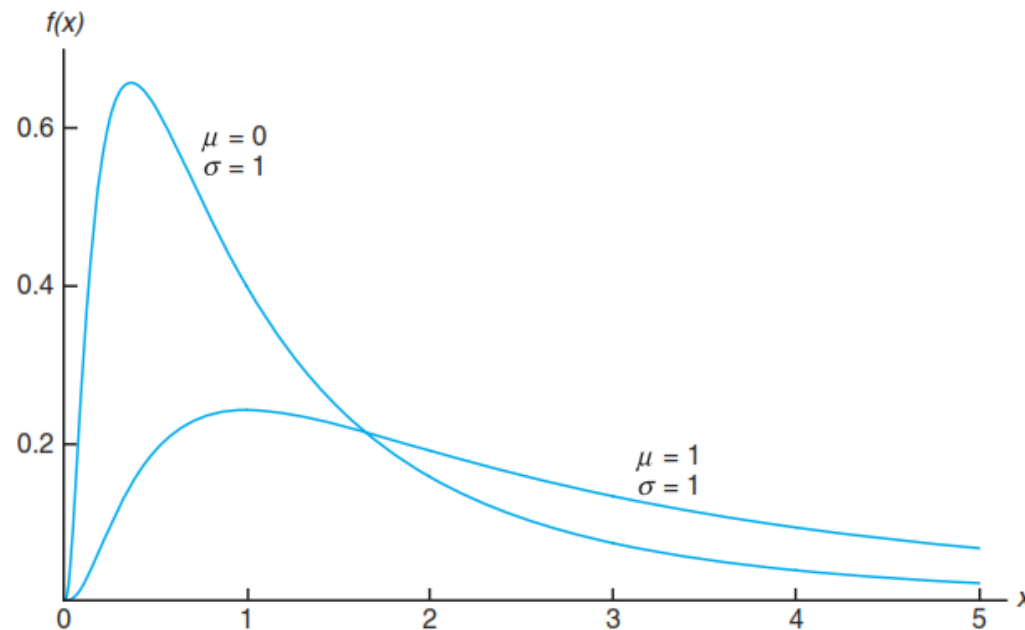


Figure 6.29: Lognormal distributions.

The graphs of the lognormal distributions are illustrated in Figure 6.29.

Theorem 6.7: The mean and variance of the lognormal distribution are

$$\mu = e^{\mu + \sigma^2/2} \text{ and } \sigma^2 = e^{2\mu + \sigma^2} (e^{\sigma^2} - 1).$$

The cumulative distribution function is quite simple due to its relationship to the normal distribution. The use of the distribution function is illustrated by the following example.

Example 6.22: Concentrations of pollutants produced by chemical plants historically are known to exhibit behavior that resembles a lognormal distribution. This is important when one considers issues regarding compliance with government regulations. Suppose it is assumed that the concentration of a certain pollutant, in parts per million, has a lognormal distribution with parameters $\mu = 3.2$ and $\sigma = 1$. What is the probability that the concentration exceeds 8 parts per million?

Solution: Let the random variable X be pollutant concentration. Then

$$P(X > 8) = 1 - P(X \leq 8).$$

Since $\ln(X)$ has a normal distribution with mean $\mu = 3.2$ and standard deviation $\sigma = 1$,

$$P(X \leq 8) = \Phi \left[\frac{\ln(8) - 3.2}{1} \right] = \Phi(-1.12) = 0.1314.$$

Here, we use Φ to denote the cumulative distribution function of the standard normal distribution. As a result, the probability that the pollutant concentration exceeds 8 parts per million is 0.1314. └



Example 6.23: The life, in thousands of miles, of a certain type of electronic control for locomotives has an approximately lognormal distribution with $\mu = 5.149$ and $\sigma = 0.737$. Find the 5th percentile of the life of such an electronic control.

Solution: From Table A.3, we know that $P(Z < -1.645) = 0.05$. Denote by X the life of such an electronic control. Since $\ln(X)$ has a normal distribution with mean $\mu = 5.149$ and $\sigma = 0.737$, the 5th percentile of X can be calculated as

$$\ln(x) = 5.149 + (0.737)(-1.645) = 3.937.$$

Hence, $x = 51.265$. This means that only 5% of the controls will have lifetimes less than 51,265 miles. └



6.10 Weibull Distribution (Optional)

Modern technology has enabled engineers to design many complicated systems whose operation and safety depend on the reliability of the various components making up the systems.

For example, a fuse may burn out, a steel column may buckle, or a heat-sensing device may fail.

Identical components subjected to identical environmental conditions will fail at different and unpredictable times.

We have seen the role that the gamma and exponential distributions play in these types of problems.

Another distribution that has been used extensively in recent years to deal with such problems is the **Weibull distribution**, introduced by the Swedish physicist Waloddi Weibull in 1939.



Weibull Distribution

The continuous random variable X has a **Weibull distribution**, with parameters α and β , if its density function is given by

$$f(x; \alpha, \beta) = \begin{cases} \alpha\beta x^{\beta-1} e^{-\alpha x^\beta}, & x > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

where $\alpha > 0$ and $\beta > 0$.

The mean and variance of the Weibull distribution are stated in the following theorem.

The reader is asked to provide the proof in Exercise 6.52 on page 206.

Theorem 6.8:

The mean and variance of the Weibull distribution are

$$\mu = \alpha^{-1/\beta} \Gamma\left(1 + \frac{1}{\beta}\right) \text{ and } \sigma^2 = \alpha^{-2/\beta} \left\{ \Gamma\left(1 + \frac{2}{\beta}\right) - \left[\Gamma\left(1 + \frac{1}{\beta}\right) \right]^2 \right\}.$$



The graphs of the Weibull distribution for $\alpha = 1$ and various values of the parameter β are illustrated in Figure 6.30.

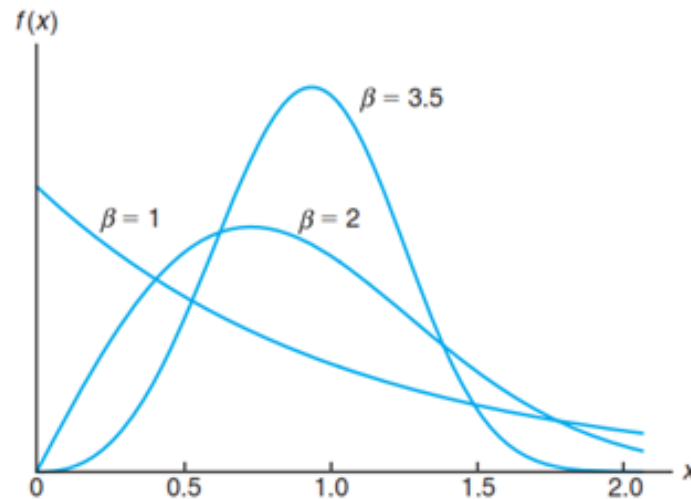


Figure 6.30: Weibull distributions ($\alpha = 1$).

We see that the curves change considerably in shape for different values of the parameter β .

If we let $\beta = 1$, the Weibull distribution reduces to the exponential distribution.

For values of $\beta > 1$, the curves become somewhat bell shaped and resemble the normal curve but display some skewness.

Like the gamma and exponential distributions, the Weibull distribution is also applied to reliability and life-testing problems such as the **time to failure** or **life length** of a component, measured from some specified time until it fails.

Let us represent this time to failure by the continuous random variable T , with probability density function $f(t)$, where $f(t)$ is the Weibull distribution.

The Weibull distribution has inherent flexibility in that it does not require the lack of memory property of the exponential distribution.

The cumulative distribution function (cdf) for the Weibull can be written in closed form and certainly is useful in computing probabilities.

cdf for Weibull Distribution

The **cumulative distribution function** for the Weibull distribution is given by

$$F(x) = 1 - e^{-\alpha x^\beta}, \quad \text{for } x \geq 0,$$

for $\alpha > 0$ and $\beta > 0$.



Example 6.24: The length of life X , in hours, of an item in a machine shop has a Weibull distribution with $\alpha = 0.01$ and $\beta = 2$. What is the probability that it fails before eight hours of usage?

Solution: $P(X < 8) = F(8) = 1 - e^{-(0.01)8^2} = 1 - 0.527 = 0.473.$



Weibull Distribution

The continuous random variable X has a **Weibull distribution**, with parameters α and β , if its density function is given by

$$f(x; \alpha, \beta) = \begin{cases} \alpha\beta x^{\beta-1} e^{-\alpha x^\beta}, & x > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

where $\alpha > 0$ and $\beta > 0$.

Gamma Distribution

The continuous random variable X has a **gamma distribution**, with parameters α and β , if its density function is given by

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, & x > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

where $\alpha > 0$ and $\beta > 0$.



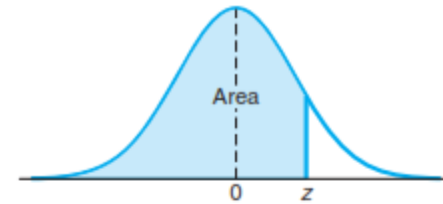


Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183



Table A.3 Areas under the Normal Curve

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

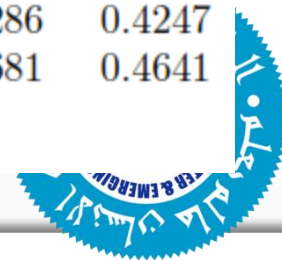


Table A.3 (continued) Areas under the Normal Curve

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767

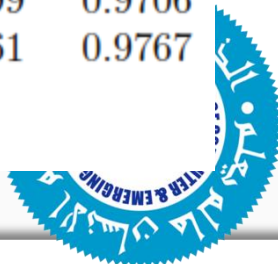


Table A.3 (continued) Areas under the Normal Curve

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998



Table A.3 Areas under the Normal Curve

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706

Table A.3 Areas under the Normal Curve

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224

Table A.3 (continued) Areas under the Normal Curve

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015

